الإمارات العربية المتحدة
وزارة التربية والتعليم

الميثاق الوطني للتعليم

FORMATIVE ASSESSMENT 2 · C2L4+5 · C3L1-4

Formative 2 Preparation

Grade 12 Advanced · Mathematics

كتيّب التجهّز للتقويم التكويني الثاني

1 2 3 4 5 6 7 8 9

9	20	60	27
objectives أهداف	marks علامة	minutes دقيقة	practice questions تدريبًا

Mr. Yousef Al Yousef · **Thursday 29 October 2026**

One page per objective · every solution written out step by step

Quiz map

9 objectives = 9 recipes · 9 أهداف = 9 وصفات

1. **The Intermediate Value Theorem** نظرية القيمة المتوسطة
   Continuous · f(a) and f(b) have opposite signs → a zero inside
2. **Continuity at a point** الاتصال عند نقطة
   Limit = value · make the two pieces meet at the break
3. **Limits at infinity and asymptotes** النهايات عند اللانهاية
   Divide by the highest power · vertical: denominator = 0 · slant: long division
4. **Average velocity** السرعة المتوسطة
   (s(b) − s(a)) ÷ (b − a) · shorter intervals for the instantaneous velocity
5. **The graph of f′** منحنى المشتقة
   f′ = 0 where f levels out · sign of f′ = rising / falling
6. **Where f is not differentiable** عدم قابلية الاشتقاق
   Absolute values: a corner where the inside is 0
7. **The power rule** قاعدة القوة
   d/dx x^r = r x^(r−1) · roots and reciprocals as powers first
8. **The product rule and tangents** قاعدة الضرب
   (fg)′ = f′g + fg′ · y − h(a) = h′(a)(x − a)
9. **The quotient rule and tangents** قاعدة القسمة
   (f/g)′ = (f′g − fg′)/g² · y − h(a) = h′(a)(x − a)

1 · Learn
Read the idea and the recipe
اقرأ الفكرة والوصفة

2 · Watch
Cover the solution, try it, then compare
غطِّ الحل وحاول ثم قارن

3 · Try
Do “Your turn”, check at the back
حلّ التدريبات وصحّح

1 The Intermediate Value Theorem
نظرية القيمة المتوسطة

A continuous function that changes sign on [a, b] has a zero in [a, b]. Compute f(a) and f(b): opposite signs.

If f is continuous on [a, b] and W lies between f(a) and f(b), then f(c) = W for some c in [a, b]. With W = 0: opposite signs at the ends mean a zero inside.

إذا كانت f متصلة على [a, b] وكانت W بين f(a) وf(b) فهناك c في [a, b] تحقق f(c) = W. وبـ W = 0: إشارتان مختلفتان تعنيان صفرًا داخل الفترة.

Recipe الوصفة

1. **Check that f is continuous (polynomials are continuous everywhere)**
   تحقّق من اتصال الدالة
2. **Compute f at both ends of the interval**
   احسب f عند طرفي الفترة
3. **Opposite signs → f(c) = 0 for some c in the interval**
   إشارتان مختلفتان ← صفر داخل الفترة
4. **Same signs → the theorem does not guarantee a zero**
   إشارة واحدة ← لا تضمن النظرية

Watch out

- Continuity must be stated: the theorem needs it.
- Same signs at the ends do NOT prove there is no zero — it only means the theorem says nothing.

Tool. f(a) · f(b) < 0 ⇒ a zero in (a, b)

Worked example Authored (shape of Theorem 4.4)

Show that $f(x) = x^3 - 2x - 5$ has a zero in the interval $[2, 3]$.

1. continuity is required
   f is a polynomial, so it is continuous on $[2, 3]$
2. negative at the left end
   $f(2) = 8 - 4 - 5 = -1 < 0$
3. positive at the right end
   $f(3) = 27 - 6 - 5 = 16 > 0$
4. the Intermediate Value Theorem with W = 0
   opposite signs: $f(c) = 0$ for some c in $[2, 3]$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Theorem 4.4) Show that $f(x) = x^3 + 3x - 1$ has a zero in the interval $[0, 1]$.
2. Authored (shape of Theorem 4.4) Show that $f(x) = \cos x - x$ has a zero in the interval $[0, 1]$ (radians).
3. Authored (shape of Theorem 4.4) Show that $x^4 - x - 3 = 0$ has a solution in the interval $[1, 2]$.

2 Continuity at a point

الاتصال عند نقطة

Lesson 2-4 · Parts 1-2 · pp. 88–89, 95
MAT.6.01.08.001 · .002

Continuous at a: the limit exists and equals f(a). To join two pieces, make their values equal where they meet.

f is continuous at x = a when lim f(x) = f(a). To remove a hole, define f(a) as the limit. For a piecewise function, make the left and right pieces meet at the break.

f متصلة عند a إذا كانت النهاية = f(a). لسدّ ثقب عرّف f(a) بالنهاية، ولدالة متعددة التعريف اجعل القطعتين تلتقيان عند نقطة الانقطاع.

Recipe الوصفة

1. **Find the limit at the point (cancel the common factor first)**
   أوجد النهاية عند النقطة
2. **Set the value at the point equal to the limit**
   ساوِ القيمة بالنهاية
3. **Piecewise: put the two formulas equal at the break x = c**
   قطعتان: ساوِ الصيغتين عند نقطة الانقطاع
4. **Solve for the constant**
   حلّ لإيجاد الثابت

Watch out

- Substitute the break value into BOTH formulas.
- Cancel the common factor before finding the limit of a 0/0 form.

Tool. lim f(x) = f(a)

Worked example Authored (shape of Example 4.2 and Ex 29-30)

Let $g(x) = \dfrac{x^2 - 4}{x - 2}$ for $x \neq 2$ and $g(2) = a$. Find a so that g is continuous at $x = 2$. Then find a so that $h(x) = \begin{cases} 2x & x < 3 \\ ax + 1 & x \geq 3 \end{cases}$ is continuous.

1. cancel x − 2, then substitute

$$\lim_{x \to 2} \frac{(x-2)(x+2)}{x-2} = \lim_{x \to 2} (x+2) = 4$$

2. continuity: the value equals the limit

$$g(2) = a = 4$$

3. the two pieces meet at x = 3

$$h: \ 2(3) = a(3) + 1$$

4. solve for a

$$6 = 3a + 1 \ \Rightarrow \ a = \tfrac{5}{3}$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Example 4.2) Let $g(x) = \dfrac{x^2 + x - 2}{x - 1}$ for $x \neq 1$ and $g(1) = a$. Find a so that g is continuous.

2. **Authored (shape of Ex 30)** Find m so that $f(x) = \begin{cases} x^2 + 1 & x \leq 2 \\ mx - 1 & x > 2 \end{cases}$ is continuous.

3. **Authored (shape of Ex 29)** A tax is $T(x) = 0.1x$ for $0 < x < 5000$ and $T(x) = c + 0.2x$ for $x \geq 5000$. Find c so that T is continuous.

3 Limits at infinity and asymptotes
النهايات عند اللانهاية وخطوط التقارب

Divide by the highest power of x. Vertical asymptote: denominator = 0. Degree one higher on top: divide for the slant asymptote.

For x → ±∞ divide the numerator and the denominator by the highest power of x (watch the sign of √x² = |x|). A slant asymptote is the quotient of the long division.

عند x → ±∞ اقسم على أعلى قوة لـ x (وانتبه إلى أن √x² = |x|). التقارب المائل هو ناتج القسمة المطولة.

Recipe الوصفة

1. **Divide top and bottom by the highest power of x**
   اقسم البسط والمقام على أعلى قوة
2. **For x → −∞: √(x²) = −x (a positive number)**
   عند −∞: √x² = −x
3. **Vertical asymptote: denominator = 0 (numerator ≠ 0)**
   التقارب الرأسي: المقام = 0
4. **Slant: numerator one degree higher → long division, quotient = y**
   المائل: القسمة المطولة

Watch out

- For x → −∞, √(x²) is −x, not x.
- Do not put the remainder into the slant asymptote.

Tool. √(x²) = |x|

Worked example Authored (shape of Ex 9-14 and Ex 29-32)

Evaluate $\lim_{x\to-\infty} \dfrac{3x}{\sqrt{x^2+9}}$.

1. x is negative, so √(x²) = −x

   for $x < 0$, $\sqrt{x^2} = -x$

2. divide top and bottom by −x

   $$\frac{3x}{\sqrt{x^2+9}} = \frac{3x}{-x\sqrt{1+9/x^2}}$$

3. the 3x ÷ (−x) gives −3

   $$= \frac{-3}{\sqrt{1+9/x^2}}$$

4. 9/x² → 0

   $$\to \frac{-3}{\sqrt{1}} = -3$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 9) Evaluate $\lim_{x\to\infty} \dfrac{3x^2+x}{5x^2-2}$.
2. Authored (shape of Ex 11) Evaluate $\lim_{x\to-\infty} \dfrac{x}{\sqrt{x^2+1}}$.
3. Authored (shape of Ex 30) Determine all vertical and slant asymptotes of $y = \dfrac{x^2-4}{x-1}$.

4 Average and instantaneous velocity
السرعة المتوسطة واللحظية

Average velocity = change in position ÷ change in time. Instantaneous velocity = what the averages approach as the interval shrinks.

v_avg = (s(b) − s(a)) / (b − a). To estimate the instantaneous velocity at t = b, find the average over shorter and shorter intervals ending at b.

السرعة المتوسطة = (s(b) − s(a)) ÷ (b − a). وللسرعة اللحظية احسب المتوسطات على فترات أقصر فأقصر تنتهي عند b.

Recipe الوصفة

1. **Find s at both times**
   أوجد s عند الزمنين
2. **Subtract the positions, divide by the time difference**
   اطرح المواضع واقسم على فرق الزمن
3. **Shorten the interval: [2.9, 3], [2.99, 3] ...**
   قصّر الفترة
4. **The averages settle on the instantaneous velocity**
   تستقر المتوسطات عند السرعة اللحظية

Watch out

- Keep the unit: ft/s.
- Position at the end minus position at the start, in that order.

Tool. v = (s(b) − s(a)) / (b − a)

Worked example Authored (shape of Ex 19-22)

The position of an object is $s(t) = 5t^2 + 3$ feet at time t seconds. Find the average velocity between $t = 1$ and $t = 3$, and estimate the instantaneous velocity at $t = 3$.

1. substitute both times

$$s(3) = 48, \quad s(1) = 8$$

2. change in position ÷ change in time

$$\frac{48 - 8}{3 - 1} = 20 \text{ ft/s}$$

3. a shorter interval ending at 3

$$[2.9, 3]: \frac{48 - 45.05}{0.1} = 29.5$$

4. the averages approach 30 ft/s

$$[2.99, 3]: \frac{48 - 47.7005}{0.01} = 29.95 \Rightarrow \approx 30 \text{ ft/s}$$

Your turn · solve in your notebook · solutions at the back

1. **Authored (shape of Ex 20)** $s(t) = 2t^3$ feet. Find the average velocity between $t = 0$ and $t = 2$.
2. **Authored (shape of Ex 21)** $s(t) = \sqrt{t^2 + 3t}$ feet. Find the average velocity between $t = 1$ and $t = 3$, to two decimal places.
3. **Authored (shape of Ex 22)** $s(t) = 4\sin(t - 1)$ feet. Find the average velocity between $t = 1$ and $t = 2$, to two decimal places (radians).

5 The graph of the derivative

منحنى المشتقة

f′ = 0 where the graph of f levels out; f′ > 0 where f rises; f′ < 0 where f falls.

Read the slope of f: horizontal tangents give zeros of f′, an increasing f gives f′ > 0, a decreasing f gives f′ < 0. Then sketch f′ through those features.

اقرأ ميل f: الماسّ الأفقي يعني f′ = 0، والدالة المتزايدة f′ > 0، والمتناقصة f′ < 0. ثم ارسم f′ عبر هذه الخصائص.

Recipe الوصفة

1. **Mark where the graph of f levels out: f′ = 0**
   علّم مواضع الماسّ الأفقي
2. **Rising parts: f′ is above the x-axis**
   الصعود: f′ فوق المحور
3. **Falling parts: f′ is below the x-axis**
   الهبوط: f′ تحت المحور
4. **Steepest places: f′ is largest in size**
   الأشد انحدارًا: |f′| الأكبر

Watch out

- f′ is the SLOPE of f, not its height.
- A turning point of f is a zero of f′ — f′ changes sign there.

Tool. slope of f = value of f′

Worked example Authored (shape of Example 2.5)

The graph of f rises for $x < 2$, falls on $(2, 4)$ and rises for $x > 4$; it levels out at $x = 2$ and $x = 4$. Describe the graph of f'.

1. horizontal tangents at x = 2 and x = 4
   $f'(2) = 0$ and $f'(4) = 0$
2. positive slope outside
   f rises on $x < 2$ and $x > 4$: $f' > 0$
3. negative slope between
   f falls on $(2, 4)$: $f' < 0$
4. above, below, above the x-axis
   f' is an upward-opening parabola-shaped curve through $(2, 0)$ and $(4, 0)$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Example 2.5) The graph of f falls everywhere, and levels out only at $x = 0$. Describe the sign of f' and where it is zero.
2. Authored (shape of Example 2.5) The graph of f is a straight line through the origin with slope -2. What is $f'(x)$ and what does its graph look like?
3. Authored (shape of Example 2.5) f falls for $x < -1$, rises on $(-1, 3)$ and falls for $x > 3$. For which values of x is $f'(x) = 0$, and where is $f' > 0$?

6 Where f is not differentiable

أين لا تكون الدالة قابلة للاشتقاق

f is NOT differentiable at a corner, a cusp, a vertical tangent or a break. Absolute values have a corner where the inside is 0.

|u| has a corner where u = 0. A sum of absolute values has a corner at each of those points: the slope jumps there.

للقيمة المطلقة |u| زاوية حيث u = 0. ومجموع قيم مطلقة له زاوية عند كل نقطة منها: الميل يقفز.

Recipe الوصفة

1. **Find where each absolute value's inside is 0**
   أوجد أين يكون داخل كل قيمة مطلقة صفرًا
2. **Write f piece by piece between those points**
   اكتب f على قطع بين هذه النقاط
3. **Compare the slopes on each side of a break point**
   قارن الميل على جانبي النقطة
4. **Different slopes → a corner → not differentiable there**
   ميلان مختلفان ← زاوية

Watch out

- Check the slope on BOTH sides — some points look like corners but are not.
- Continuity is not enough: a corner is continuous but not differentiable.

Tool. corner: the slope jumps

Worked example Authored (shape of Ex 29)

At which x-values is $f(x) = |x - 1| + |x + 2|$ not differentiable?

1. the inside of each absolute value

   $x - 1 = 0 \Rightarrow x = 1; \quad x + 2 = 0 \Rightarrow x = -2$

2. both absolute values change sign here

   $x < -2: \ f = -2x - 1, \text{ slope } -2$

3. a constant piece

   $-2 < x < 1: \ f = 3, \text{ slope } 0$

4. the slope changes at both points

   $x > 1: \ f = 2x + 1, \text{ slope } 2 \Rightarrow \text{ corners at } x = -2 \text{ and } x = 1$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 29) At which x-values is $f(x) = |x| + |x - 3|$ not differentiable?
2. Authored (shape of Ex 29) At which x-values is $f(x) = |x^2 - 1|$ not differentiable?
3. Authored (shape of Ex 29) At which x-values is $f(x) = |x + 1| - |x|$ not differentiable?

7 The power rule

قاعدة القوة

d/dx x^r = r · x^(r – 1) for any real r. Write roots and fractions as powers FIRST.

Bring the exponent down as a factor, then lower the exponent by 1. This works for fractional and negative exponents too.

انزل الأسّ كمعامل ثم انقص منه واحدًا. تصلح القاعدة للأسس الكسرية والسالبة.

Recipe الوصفة

1. **Rewrite roots and reciprocals as powers of x**
   اكتب الجذور والمقلوبات كقوى لـ x
2. **Bring the exponent r down in front**
   أنزل الأسّ r
3. **Subtract 1 from the exponent**
   اطرح 1 من الأسّ
4. **Simplify the new exponent**
   بسّط الأسّ

Watch out

- r – 1 with a fraction: 7/2 – 1 = 5/2.
- A negative exponent stays negative: –5/2 – 1 = –7/2.

Tool. d/dx x^r = r x^(r–1)

Worked example Authored (shape of Theorem 3.2)

Find $\frac{d}{dx}x^{7/2}$.

1. the exponent is r

$r = \frac{7}{2}$

2. bring r down, subtract 1 from the exponent

$\frac{d}{dx}x^{7/2} = \frac{7}{2}x^{7/2-1}$

3. 7/2 – 1 = 5/2

$= \frac{7}{2}x^{5/2}$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}x^{-5/2}$.
2. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}\sqrt{x^3}$.
3. Authored (shape of Theorem 3.2) Find $\frac{d}{dx}\frac{1}{x^4}$.

8 The product rule and tangent lines
قاعدة الضرب والمماسات

(fg)′ = f′g + fg′. A tangent line needs the point AND the slope: y − h(a) = h′(a)(x − a).

Differentiate a product with (fg)′ = f′g + fg′. For a tangent line at x = a find h(a) and h′(a), then write y − h(a) = h′(a)(x − a).

مشتقة الضرب (fg)′ = f′g + fg′. والمماس عند a: y − h(a) = h′(a)(x − a) بعد حساب h(a) وh′(a).

Recipe الوصفة

1. **Write h = f·g and find h(a) = f(a)g(a)**
   احسب h(a) = f(a)g(a)
2. **h′(a) = f′(a)g(a) + f(a)g′(a)**
   h′(a) = f′g + fg′
3. **Substitute the given values carefully**
   عوّض القيم المعطاة
4. **y − h(a) = h′(a)(x − a), then simplify**
   اكتب معادلة المماس وبسّطها

Watch out

- Both terms of the product rule: f′g AND fg′.
- The tangent line passes through (a, h(a)), not (a, 0).

Tool. (fg)′ = f′g + fg′

Worked example Authored (shape of Ex 17-18 and 21)

Assume $f(2) = 3$, $f'(2) = -1$ and $g(2) = 4$, $g'(2) = 2$. Find the tangent line to $y = h(x)$ at $x = 2$, where $h(x) = f(x)g(x)$.

1. the point on the curve

$$h(2) = f(2)g(2) = 3 \cdot 4 = 12$$

2. the product rule

$$h'(2) = f'(2)g(2) + f(2)g'(2) = (-1)(4) + (3)(2) = 2$$

3. point-slope form

$$y - 12 = 2(x - 2)$$

4. simplify

$$y = 2x + 8$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 17) Find the tangent line to $f(x) = (x^3 + 2)(x - 1)$ at $x = 2$.
2. Authored (shape of Ex 21) Assume $f(1) = 2$, $f'(1) = 3$ and $g(1) = -1$, $g'(1) = 4$. Find the tangent line to $h = fg$ at $x = 1$.
3. Authored (shape of Ex 18) Find the tangent line to $f(x) = (x^2 - 3)(2x + 1)$ at $x = 1$.

9 The quotient rule and tangent lines
قاعدة القسمة والمماسات

(f/g)′ = (f′g − fg′) / g². The numerator order matters: f′g FIRST, minus fg′.

Differentiate a quotient with (f/g)′ = (f′g − fg′)/g². For a tangent line at x = a use h(a), h′(a) and y − h(a) = h′(a)(x − a).

مشتقة القسمة (f/g)′ = (f′g − fg′) ÷ g². والمماس عند a: y − h(a) = h′(a)(x − a).

Recipe الوصفة

1. **Write h = f/g and find h(a) = f(a)/g(a)**
   احسب h(a) = f(a)/g(a)
2. **h′(a) = (f′g − fg′)/g² at a**
   h′(a) = (f′g − fg′)/g²
3. **Substitute the values, keep the order of the subtraction**
   عوّض مع الحفاظ على ترتيب الطرح
4. **y − h(a) = h′(a)(x − a), then simplify**
   اكتب المماس وبسّطه

Watch out

- Divide by g SQUARED, not by g.
- The order is f′g − fg′; swapping them changes the sign.

Tool. (f/g)′ = (f′g − fg′) / g²

Worked example Authored (shape of Ex 19-20 and 22)

Assume $f(0) = 2$, $f'(0) = -3$ and $g(0) = 1$, $g'(0) = 2$. Find the tangent line to $y = h(x)$ at $x = 0$, where $h(x) = \dfrac{f(x)}{g(x)}$.

1. the point on the curve

$$h(0) = \frac{f(0)}{g(0)} = \frac{2}{1} = 2$$

2. the quotient rule

$$h'(0) = \frac{f'(0)g(0) - f(0)g'(0)}{[g(0)]^2} = \frac{(-3)(1) - (2)(2)}{1} = -7$$

3. point-slope form

$$y - 2 = -7(x - 0) \Rightarrow y = -7x + 2$$

Your turn · solve in your notebook · solutions at the back

1. Authored (shape of Ex 19) Find the tangent line to $f(x) = \dfrac{x+2}{x^2+3}$ at $x = 1$.
2. Authored (shape of Ex 22) Assume $f(1) = 4$, $f'(1) = 2$ and $g(1) = 2$, $g'(1) = -1$. Find the tangent line to $h = \dfrac{f}{g}$ at $x = 1$.
3. Authored (shape of Ex 20) Find the tangent line to $f(x) = \dfrac{3x-1}{x+1}$ at $x = 0$.

Objective 1 · The Intermediate Value Theorem نظرية القيمة المتوسطة

1.1 Authored (shape of Theorem 4.4)

Show that $f(x) = x^3 + 3x - 1$ has a zero in the interval $[0, 1]$.

1. continuity
   f is continuous (a polynomial)
2. opposite signs
   $f(0) = -1 < 0$ and $f(1) = 1 + 3 - 1 = 3 > 0$
3. the Intermediate Value Theorem
   so $f(c) = 0$ for some c in $[0, 1]$

1.2 Authored (shape of Theorem 4.4)

Show that $f(x) = \cos x - x$ has a zero in the interval $[0, 1]$ (radians).

1. continuity
   f is continuous (cos and x are)
2. opposite signs
   $f(0) = 1 > 0$ and $f(1) = \cos 1 - 1 \approx -0.46 < 0$
3. the Intermediate Value Theorem
   so $f(c) = 0$ for some c in $[0, 1]$

1.3 Authored (shape of Theorem 4.4)

Show that $x^4 - x - 3 = 0$ has a solution in the interval $[1, 2]$.

1. continuity
   $f(x) = x^4 - x - 3$ is continuous (a polynomial)
2. opposite signs
   $f(1) = 1 - 1 - 3 = -3 < 0$ and $f(2) = 16 - 2 - 3 = 11 > 0$
3. the Intermediate Value Theorem
   so $f(c) = 0$ for some c in $[1, 2]$

Objective 2 · Continuity at a point — الاتصال عند نقطة

2.1 Authored (shape of Example 4.2)

Let $g(x) = \dfrac{x^2 + x - 2}{x - 1}$ for $x \neq 1$ and $g(1) = a$. Find a so that g is continuous.

1. factor, cancel x − 1

$$\lim_{x \to 1} \frac{(x+2)(x-1)}{x-1} = \lim_{x \to 1} (x+2) = 3$$

2. the value equals the limit

$a = 3$

2.2 Authored (shape of Ex 30)

Find m so that $f(x) = \begin{cases} x^2 + 1 & x \le 2 \\ mx - 1 & x > 2 \end{cases}$ is continuous.

1. the pieces meet at x = 2

$x = 2 : \; 2^2 + 1 = 2m - 1$

2. solve for m

$5 = 2m - 1 \;\Rightarrow\; m = 3$

2.3 Authored (shape of Ex 29)

A tax is $T(x) = 0.1x$ for $0 < x < 5000$ and $T(x) = c + 0.2x$ for $x \ge 5000$. Find c so that T is continuous.

1. the pieces meet at x = 5000

$x = 5000 : \; 0.1(5000) = c + 0.2(5000)$

2. solve for c

$500 = c + 1000 \;\Rightarrow\; c = -500$

Objective 3 · Limits at infinity and asymptotes — النهايات عند اللانهاية وخطوط التقارب

3.1 Authored (shape of Ex 9)

Evaluate $\lim_{x \to \infty} \dfrac{3x^2 + x}{5x^2 - 2}$.

1. divide by x²

$$\frac{3x^2 + x}{5x^2 - 2} \cdot \frac{1/x^2}{1/x^2} = \frac{3 + 1/x}{5 - 2/x^2}$$

2. 1/x → 0 and 1/x² → 0

$\to \dfrac{3}{5}$

3.2 Authored (shape of Ex 11)

Evaluate $\lim_{x\to-\infty} \dfrac{x}{\sqrt{x^2+1}}$.

1. x is negative

   for $x<0$, $\sqrt{x^2+1} = -x\sqrt{1+1/x^2}$

2. the limit is −1

   $$\frac{x}{-x\sqrt{1+1/x^2}} = \frac{-1}{\sqrt{1+1/x^2}} \to -1$$

3.3 Authored (shape of Ex 30)

Determine all vertical and slant asymptotes of $y = \dfrac{x^2-4}{x-1}$.

1. denominator = 0, numerator ≠ 0

   $x-1=0$ and $1^2-4=-3 \neq 0$: vertical asymptote $x=1$

2. long division

   $$\frac{x^2-4}{x-1} = x+1-\frac{3}{x-1} \Rightarrow \text{slant asymptote } y=x+1$$

Objective 4 · Average and instantaneous velocity — السرعة المتوسطة واللحظية

4.1 Authored (shape of Ex 20)

$s(t)=2t^3$ **feet. Find the average velocity between** $t=0$ **and** $t=2$.

1. substitute both times

   $s(2)=16, \quad s(0)=0$

2. change in position ÷ change in time

   $$\frac{16-0}{2-0} = 8 \text{ ft/s}$$

4.2 Authored (shape of Ex 21)

$s(t)=\sqrt{t^2+3t}$ **feet. Find the average velocity between** $t=1$ **and** $t=3$**, to two decimal places.**

1. substitute both times

   $s(3)=\sqrt{18}\approx 4.243, \quad s(1)=\sqrt{4}=2$

2. change in position ÷ change in time

   $$\frac{4.243-2}{3-1} \approx 1.12 \text{ ft/s}$$

4.3 Authored (shape of Ex 22)

$s(t) = 4\sin(t-1)$ feet. Find the average velocity between $t = 1$ and $t = 2$, to two decimal places (radians).

1. substitute both times

$$s(2) = 4\sin 1 \approx 3.366, \quad s(1) = 4\sin 0 = 0$$

2. change in position ÷ change in time

$$\frac{3.366 - 0}{2 - 1} \approx 3.37 \text{ ft/s}$$

Objective 5 · The graph of the derivative

منحنى المشتقة

5.1 Authored (shape of Example 2.5)

The graph of f falls everywhere, and levels out only at $x = 0$. Describe the sign of f' and where it is zero.

1. no rising part

f falls : $f' \le 0$ everywhere

2. the only level point

$f'(0) = 0$: the graph of f' touches the x-axis at $x = 0$

5.2 Authored (shape of Example 2.5)

The graph of f is a straight line through the origin with slope -2. What is $f'(x)$ and what does its graph look like?

1. the slope never changes

$f(x) = -2x \Rightarrow f'(x) = -2$

2. constant derivative

a horizontal line at $y = -2$

5.3 Authored (shape of Example 2.5)

f falls for $x < -1$, rises on $(-1, 3)$ and falls for $x > 3$. For which values of x is $f'(x) = 0$, and where is $f' > 0$?

1. f levels out at the turning points

$f'(-1) = 0$ and $f'(3) = 0$

2. rising means a positive slope

$f' > 0$ on $(-1, 3)$ (where f rises)

Objective 6 · Where f is not differentiable — أين لا تكون الدالة قابلة للاشتقاق

6.1 Authored (shape of Ex 29)

At which x-values is $f(x) = |x| + |x-3|$ not differentiable?

1. the inside of each absolute value

   $x = 0$ and $x = 3$ make the insides zero

2. the slope changes at both points

   slopes $-2,\ 0,\ 2 \Rightarrow$ not differentiable at $x = 0$ and $x = 3$

6.2 Authored (shape of Ex 29)

At which x-values is $f(x) = |x^2 - 1|$ not differentiable?

1. the inside is zero

   $x^2 - 1 = 0 \Rightarrow x = \pm 1$

2. a corner in the graph

   the slope of $|x^2 - 1|$ changes from -2 to 2 at $x = 1 \Rightarrow$ not differentiable at $x = \pm 1$

6.3 Authored (shape of Ex 29)

At which x-values is $f(x) = |x+1| - |x|$ not differentiable?

1. the inside of each absolute value

   $x = -1$ and $x = 0$ make the insides zero

2. the three pieces

   $x < -1 :\ f = -(x+1) + x = -1;\ -1 < x < 0 :\ f = x + 1 + x = 2x + 1;\ x >$

3. the slope changes at both points

   slopes $0,\ 2,\ 0 \Rightarrow$ not differentiable at $x = -1$ and $x = 0$

Objective 7 · The power rule — قاعدة القوة

7.1 Authored (shape of Theorem 3.2)

Find $\frac{d}{dx} x^{-5/2}$.

1. $r = -5/2$

   $$\frac{d}{dx} x^{-5/2} = -\frac{5}{2}\, x^{-5/2-1} = -\frac{5}{2}\, x^{-7/2}$$

7.2 Authored (shape of Theorem 3.2)

Find $\frac{d}{dx}\sqrt{x^3}$.

1. write the root as a power

$\sqrt{x^3} = x^{3/2}$

2. bring 3/2 down, subtract 1

$\frac{d}{dx}x^{3/2} = \frac{3}{2}x^{1/2}$

7.3 Authored (shape of Theorem 3.2)

Find $\frac{d}{dx}\frac{1}{x^4}$.

1. write the reciprocal as a power

$\frac{1}{x^4} = x^{-4}$

2. bring −4 down, subtract 1

$\frac{d}{dx}x^{-4} = -4x^{-5}$

Objective 8 · The product rule and tangent lines قاعدة الضرب والمماسات

8.1 Authored (shape of Ex 17)

Find the tangent line to $f(x) = (x^3 + 2)(x - 1)$ **at** $x = 2$.

1. the point on the curve

$f(2) = (10)(1) = 10$

2. the product rule

$f'(x) = 3x^2(x - 1) + (x^3 + 2)(1) \Rightarrow f'(2) = 12 + 10 = 22$

3. point-slope form

$y - 10 = 22(x - 2) \Rightarrow y = 22x - 34$

8.2 Authored (shape of Ex 21)

Assume $f(1) = 2$, $f'(1) = 3$ **and** $g(1) = -1$, $g'(1) = 4$. **Find the tangent line to** $h = fg$ **at** $x = 1$.

1. the point on the curve

$h(1) = (2)(-1) = -2$

2. the product rule

$h'(1) = 3(-1) + (2)(4) = 5$

3. point-slope form

$y + 2 = 5(x - 1) \Rightarrow y = 5x - 7$

8.3 Authored (shape of Ex 18)

Find the tangent line to $f(x) = (x^2 - 3)(2x + 1)$ at $x = 1$.

1. the point on the curve

$$f(1) = (-2)(3) = -6$$

2. the product rule

$$f'(x) = 2x(2x + 1) + (x^2 - 3)(2) \Rightarrow f'(1) = 6 - 4 = 2$$

3. point-slope form

$$y + 6 = 2(x - 1) \Rightarrow y = 2x - 8$$

Objective 9 · The quotient rule and tangent lines

قاعدة القسمة والمماسات

9.1 Authored (shape of Ex 19)

Find the tangent line to $f(x) = \dfrac{x + 2}{x^2 + 3}$ at $x = 1$.

1. the point on the curve

$$f(1) = \frac{3}{4}$$

2. the quotient rule

$$f'(x) = \frac{(1)(x^2 + 3) - (x + 2)(2x)}{(x^2 + 3)^2} \Rightarrow f'(1) = \frac{4 - 6}{16} = -\frac{1}{8}$$

3. point-slope form

$$y - \tfrac{3}{4} = -\tfrac{1}{8}(x - 1) \Rightarrow y = -\tfrac{1}{8}x + \tfrac{7}{8}$$

9.2 Authored (shape of Ex 22)

Assume $f(1) = 4$, $f'(1) = 2$ and $g(1) = 2$, $g'(1) = -1$. Find the tangent line to $h = \dfrac{f}{g}$ at $x = 1$.

1. the point on the curve

$$h(1) = \frac{4}{2} = 2$$

2. the quotient rule

$$h'(1) = \frac{(2)(2) - (4)(-1)}{2^2} = \frac{8}{4} = 2$$

3. point-slope form

$$y - 2 = 2(x - 1) \Rightarrow y = 2x$$

9.3 Authored (shape of Ex 20)

Find the tangent line to $f(x) = \dfrac{3x-1}{x+1}$ at $x = 0$.

1. the point on the curve

$$f(0) = \frac{-1}{1} = -1$$

2. the quotient rule

$$f'(x) = \frac{3(x+1) - (3x-1)(1)}{(x+1)^2} = \frac{4}{(x+1)^2} \Rightarrow f'(0) = 4$$

3. point-slope form

$$y + 1 = 4x \Rightarrow y = 4x - 1$$

Ready?

Tick each line · ضع علامة

- [ ] **I can use the Intermediate Value Theorem**
  أستطيع استخدام نظرية القيمة المتوسطة
- [ ] **I can make a function continuous at a point**
  أستطيع جعل الدالة متصلة عند نقطة
- [ ] **I can find a limit at infinity and the asymptotes**
  أستطيع إيجاد نهاية عند اللانهاية وخطوط التقارب
- [ ] **I can find average and instantaneous velocity**
  أستطيع إيجاد السرعة المتوسطة واللحظية
- [ ] **I can read the graph of f′ from the graph of f**
  أستطيع قراءة منحنى f′ من منحنى f
- [ ] **I can say where a function is not differentiable**
  أستطيع تحديد نقاط عدم القابلية للاشتقاق
- [ ] **I can use the power rule**
  أستطيع استخدام قاعدة القوة
- [ ] **I can find a tangent line with the product rule**
  أستطيع إيجاد المماس بقاعدة الضرب
- [ ] **I can find a tangent line with the quotient rule**
  أستطيع إيجاد المماس بقاعدة القسمة

On the day

Nine questions are on the computer and three are on paper. Write down the rule or the theorem you use before you calculate.

Every question

On paper, show every line: the first step already earns a mark.

One mark for every written step

لكل خطوة تكتبها علامة